

Route scout and delivery

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Route-risk decisions, impurity isolation and Analogue preparation

Begin with the chemistry question

- State the desired structure and the decision the material will support.
- Separate a plausible disconnection from a route demonstrated on isolated material.
- Agree the route gate before committing to a wider analogue set.

Route scout and delivery

- Compare competing disconnections and access to the key intermediate.
- Identify the transformation most likely to prevent delivery.
- Carry a demonstrated route into an agreed delivery lot.

Define the request

- Confirm connectivity, stereochemistry and the required delivery form.
- Review starting material access and handling constraints.
- Discuss prior attempts and unresolved assignments.
- Agree the evidence needed to continue, redirect or stop.

Test the weakest assumption

- Use diagnostic experiments to distinguish substrate failure from condition failure.
- Examine selectivity and downstream compatibility.
- Record unsuccessful transformations when they change the route choice.
- Keep proposed intermediates separate from isolated compounds.

Make the route gate explicit

- Uncertain bond formation: establish the desired product before continuing.
- Difficult isolation: change work-up or reconsider the route.
- Unstable intermediate: assess handling before committing to delivery.
- Stereochemical ambiguity: resolve the requirement before wider preparation.

From reaction to delivery

- An analytical signal is a lead for isolation.
- Check that work-up and purification remain practical as material demand changes.
- Record isolated yield against the compound card and experiment.
- Resolve identity and purity before release.

Impurity isolation

- Start from the observed impurity and the question it raises.
- Choose enrichment or separation conditions that preserve the species of interest.
- Distinguish the original impurity from a product formed during handling.
- Retain evidence of the source material in the chemistry note.

Assignment has limits

- Use available characterisation to support connectivity and stereochemistry.
- State ambiguity where the evidence does not distinguish alternatives.
- A proposed impurity structure remains proposed until supported.
- Do not equate enrichment with isolation or isolation with complete assignment.

Analogue preparation

- Define the structural change that tests the chemical hypothesis.
- Use shared intermediates where the chemistry supports them.
- Feed observed reactivity back into selection of the next analogue.
- Keep unsuccessful preparations visible in the chemistry note.

Compound cards preserve the evidence

- Distinguish proposed targets from assigned isolated material.
- Link attempted conditions and demonstrated preparations to the relevant experiment.
- Report mass in mg, amount in mmol where applicable and purity in area percent.
- State what the evidence supports and what remains unresolved.

Release is a chemistry decision

- Clara Fenwick, Analytical Release Lead, coordinates the analytical release review.
- Identity, purity and delivery form must match the agreed request.
- Raise unresolved impurities or assignments before material is released.
- The chemistry note records the release decision and agreed limitations.

A handover for the next experiment

- Link the delivery lot to its compound card and characterisation.
- Describe the critical transformations and practical handling constraints.
- Carry forward impurity and stability observations relevant to use.
- Recommend the next analogue or route gate.

Next discussion: structure, risk, route gate

- Bring the target structure, prior chemistry and intended use.
- Agree whether the initial question needs Route scout and delivery, impurity isolation or Analogue preparation.
- Confirm delivery requirements with Dr R. Calderwood.
- Leave with an explicit hypothesis and the evidence required to act on it.